Mathematical Methods 1

Roger D.

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Chapter 1

Vectors

This chapter follows the lecturer's vector algebra, including the **index-notation**. It provides a foundation for later vector calculus. Bold letters replace the manuscript's underlined vector notation. All coordinates refer to a right-handed orthonormal Cartesian system.

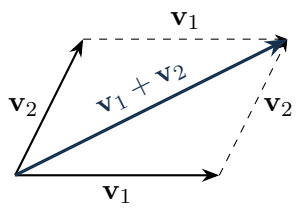
1.1 A geometric approach to vectors

Informally, a vector is an arrow with a direction and a length. More precisely, a geometric vector is an equivalence class of directed line segments in three-dimensional Euclidean space. The segment from A to B represents $\overrightarrow{AB}$; A is its initial point and B its terminal point. Two directed segments represent the same vector when they have the same length and direction, even if their starting points differ. Parallel arrows pointing in opposite directions do not represent the same nonzero vector.

Examples from mechanics are the position vector $\mathbf{r} = \overrightarrow{OP}$ of a point relative to a fixed origin O, and its velocity $\mathbf{v}$. If the endpoints coincide, $\overrightarrow{AA} = \mathbf{0}$ is the zero vector, which has zero magnitude and no distinguished direction.

1.1.1 Addition, subtraction and scalar multiplication

If $\mathbf{v}_1$ and $\mathbf{v}_2$ are successive displacements, their total is $\mathbf{v}_1 + \mathbf{v}_2$. Place the tail of the second arrow at the tip of the first; the resultant joins the initial tail to the final tip. The parallelogram construction shows $\mathbf{v}_1 + \mathbf{v}_2 = \mathbf{v}_2 + \mathbf{v}_1$.



Subtraction reverses addition: if $\mathbf{a} = \mathbf{b} + \mathbf{c}$, then $\mathbf{c} = \mathbf{a} - \mathbf{b}$. In particular, $-\mathbf{v}$ has the same magnitude as $\mathbf{v}$ and the opposite direction, and $\mathbf{v}+(-\mathbf{v}) = \mathbf{0}$. Repeated addition gives $2\mathbf{v} = \mathbf{v}+\mathbf{v}$ and $3\mathbf{v} = \mathbf{v} + \mathbf{v} + \mathbf{v}$. For $\lambda \in \mathbb{R}$, the vector $\lambda\mathbf{v}$ has length $|\lambda|\,|\mathbf{v}|$; for nonzero $\mathbf{v}$ it points in the same direction when $\lambda > 0$ and in the opposite direction when $\lambda < 0$. Also $0\mathbf{v} = \mathbf{0}$.

1.1.2 Properties of the operations

For vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ and real numbers λ, μ,

$$\begin{aligned} \mathbf{v}+\mathbf{w} &= \mathbf{w}+\mathbf{v}, & (\mathbf{u}+\mathbf{v})+\mathbf{w} &= \mathbf{u}+(\mathbf{v}+\mathbf{w}),\\ \mathbf{v}+\mathbf{0} &= \mathbf{v}, & \mathbf{v}+(-\mathbf{v}) &= \mathbf{0},\\ (\lambda+\mu)\mathbf{v} &= \lambda\mathbf{v}+\mu\mathbf{v}, & \lambda(\mathbf{v}+\mathbf{w}) &= \lambda\mathbf{v}+\lambda\mathbf{w},\\ \lambda(\mu\mathbf{v}) &= (\lambda\mu)\mathbf{v}, & 1\mathbf{v} &= \mathbf{v}. \end{aligned}$$

These can be verified geometrically. Matrices, functions and polynomials also obey these vector-space rules with suitable addition and scalar multiplication; geometric arrows are one instance of a broader algebraic idea.

Example 1.1 (Lecture Example 1: the midpoint theorem)**.** Let ABC be a triangle, and let D, E be the midpoints of AB, BC respectively. Show that DE is parallel to AC and has half its length.

$$\begin{aligned} \overrightarrow{DE} &= \overrightarrow{DB}+\overrightarrow{BE}\\ &= \frac{1}{2}\overrightarrow{AB}+\frac{1}{2}\overrightarrow{BC}\\ &= \frac{1}{2}(\overrightarrow{AB}+\overrightarrow{BC}) = \frac{1}{2}\overrightarrow{AC}. \end{aligned}$$

The positive scalar $1/2$ gives the claimed direction and length.

1.2 Cartesian coordinates

Fix an origin O and perpendicular coordinate axes x, y, z in a right-handed orientation. Let $\mathbf{i}, \mathbf{j}, \mathbf{k}$ be the unit vectors along these axes. A point $P = (x, y, z)$ has position vector

$$\mathbf{r} = \overrightarrow{OP} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}.$$

Conversely, translate the tail of any vector to O to write it uniquely as

$$\mathbf{v} = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \langle v_1, v_2, v_3 \rangle.$$

The numbers v_1, v_2, v_3 are its components. Vector operations become

$$\begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} + \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} u_1+v_1 \\ u_2+v_2 \\ u_3+v_3 \end{pmatrix}, \qquad \lambda \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} \lambda v_1 \\ \lambda v_2 \\ \lambda v_3 \end{pmatrix}.$$

Example 1.2 (Lecture Example 2: coordinates of a midpoint)**.** Let $A = (a_1, a_2, a_3)$ and $B = (b_1, b_2, b_3)$, and let M be their midpoint.

$$\begin{aligned} \overrightarrow{OM} &= \overrightarrow{OA}+\frac{1}{2}\overrightarrow{AB}\\ &= \overrightarrow{OA}+\frac{1}{2}(\overrightarrow{OB}-\overrightarrow{OA})\\ &= \frac{1}{2}(\overrightarrow{OA}+\overrightarrow{OB}). \end{aligned}$$

Thus

$$M = \left(\frac{a_1+b_1}{2}, \frac{a_2+b_2}{2}, \frac{a_3+b_3}{2} \right).$$

1.2.1 Length and unit vectors

Pythagoras, applied first in a coordinate plane and then in space, gives

$$|\mathbf{v}|^2 = v_1^2 + v_2^2 + v_3^2, \qquad |\mathbf{v}| = \sqrt{v_1^2 + v_2^2 + v_3^2}.$$

Since $|\lambda\mathbf{v}| = |\lambda|\,|\mathbf{v}|$, every nonzero vector can be normalised:

$$\widehat{\mathbf{v}} = \frac{\mathbf{v}}{|\mathbf{v}|}, \qquad |\widehat{\mathbf{v}}| = \frac{|\mathbf{v}|}{|\mathbf{v}|} = 1.$$

A vector of length 1 is called a unit vector. The zero vector cannot be normalised.

1.3 Straight lines

A line is specified by a point P_0 on it and a nonzero direction vector $\mathbf{v}$. If $\mathbf{r}_0 = \overrightarrow{OP_0}$, then its points satisfy

$$\boxed{\mathbf{r} = \mathbf{r}_0 + t\mathbf{v}, \qquad t \in \mathbb{R}.}$$

Writing $P_0 = (x_0, y_0, z_0)$ and $\mathbf{v} = \langle v_1, v_2, v_3\rangle$ gives the parametric equations

$$x = x_0 + tv_1, \qquad y = y_0 + tv_2, \qquad z = z_0 + tv_3.$$

If all three direction components are nonzero, eliminate t to obtain

$$\frac{x - x_0}{v_1} = \frac{y - y_0}{v_2} = \frac{z - z_0}{v_3}.$$

If a component is zero, use the corresponding constant-coordinate equation instead of dividing by zero.

Example 1.3 (Lecture Example 3: a line through two points)**.** Find the parametric and Cartesian forms of the line through $P_0 = (1, 2, 3)$ and $Q_0 = (2, 1, -1)$. Its direction is

$$\overrightarrow{P_0Q_0} = \overrightarrow{OQ_0} - \overrightarrow{OP_0} = \langle 2, 1, -1\rangle - \langle 1, 2, 3\rangle = \langle 1, -1, -4\rangle.$$

Thus

$$\mathbf{r} = \langle 1, 2, 3\rangle + t\langle 1, -1, -4\rangle,$$

so

$$x = 1 + t, \qquad y = 2 - t, \qquad z = 3 - 4t, \qquad \frac{x-1}{1} = \frac{y-2}{-1} = \frac{z-3}{-4}.$$

1.4 The scalar or dot product

The mechanical motivation is work done by a constant force $\mathbf{F}$ through a displacement $\mathbf{d}$: $W = |\mathbf{F}||\mathbf{d}|\cos\theta$. For nonzero vectors meeting at angle $\theta \in [0, \pi]$, define

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}|\cos\theta.$$

Set the product to zero if either vector is zero. Two nonzero vectors are orthogonal exactly when $\theta = \pi/2$, equivalently $\mathbf{a} \cdot \mathbf{b} = 0$.

1.4.1 Deriving the component formula

The cosine rule gives

$$|\mathbf{a}-\mathbf{b}|^2 = |\mathbf{a}|^2 + |\mathbf{b}|^2 - 2|\mathbf{a}||\mathbf{b}|\cos\theta.$$

Therefore

$$\begin{aligned}\mathbf{a}\cdot\mathbf{b} &= \frac{1}{2}(|\mathbf{a}|^2 + |\mathbf{b}|^2 - |\mathbf{a}-\mathbf{b}|^2)\\ &= \frac{1}{2}\sum_{i=1}^{3}(a_i^2 + b_i^2 - (a_i-b_i)^2)\\ &= a_1b_1 + a_2b_2 + a_3b_3.\end{aligned}$$

In particular,

$$\boxed{\mathbf{a}\cdot\mathbf{b} = |\mathbf{a}||\mathbf{b}|\cos\theta = a_1b_1 + a_2b_2 + a_3b_3}, \qquad \theta = \arccos\frac{\mathbf{a}\cdot\mathbf{b}}{|\mathbf{a}||\mathbf{b}|}.$$

The angle formula requires both vectors to be nonzero.

1.4.2 Properties

For real λ,

$$\begin{aligned}\mathbf{a}\cdot(\mathbf{b}+\mathbf{c}) &= \mathbf{a}\cdot\mathbf{b} + \mathbf{a}\cdot\mathbf{c},\\ (\lambda\mathbf{a})\cdot\mathbf{b} &= \lambda(\mathbf{a}\cdot\mathbf{b}),\\ \mathbf{a}\cdot\mathbf{b} &= \mathbf{b}\cdot\mathbf{a}.\end{aligned}$$

For example, the first identity follows by expanding components:

$$\begin{aligned}\mathbf{a}\cdot(\mathbf{b}+\mathbf{c}) &= a_1(b_1+c_1) + a_2(b_2+c_2) + a_3(b_3+c_3)\\ &= (a_1b_1 + a_2b_2 + a_3b_3) + (a_1c_1 + a_2c_2 + a_3c_3).\end{aligned}$$

The other two follow similarly.

1.4.3 Equations of planes

A plane through P_0 with a nonzero normal vector $\mathbf{n}$ consists of points P for which $\overrightarrow{P_0P} \perp \mathbf{n}$. Writing their position vectors as $\mathbf{r}_0, \mathbf{r}$,

$$\begin{aligned}(\mathbf{r}-\mathbf{r}_0)\cdot\mathbf{n} &= 0,\\ \mathbf{r}\cdot\mathbf{n} &= \mathbf{r}_0\cdot\mathbf{n} = d.\end{aligned}$$

If $\mathbf{n} = \langle n_1, n_2, n_3\rangle$, the **Cartesian equation** is

$$\boxed{n_1x + n_2y + n_3z = d}.$$

Alternatively, with two nonparallel directions $\mathbf{u}, \mathbf{v}$ parallel to the plane,

$$\boxed{\mathbf{r} = \mathbf{r}_0 + s\mathbf{u} + t\mathbf{v}, \qquad s,t\in\mathbb{R}.}$$

The nonparallel condition ensures that this describes a plane rather than a line.

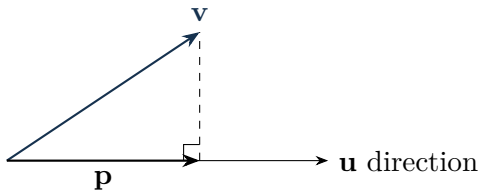
1.4.4 Orthogonal projection

For $\mathbf{u} \neq \mathbf{0}$, project $\mathbf{v}$ onto the direction of $\mathbf{u}$. The signed component is $|\mathbf{v}|\cos\theta$, and hence

$$\begin{aligned}\mathbf{p} = \operatorname{proj}_{\mathbf{u}} \mathbf{v} &= |\mathbf{v}|\cos\theta \frac{\mathbf{u}}{|\mathbf{u}|} \\ &= \boxed{\frac{\mathbf{v}\cdot\mathbf{u}}{|\mathbf{u}|^2}\mathbf{u}}.\end{aligned}$$

Check that the remaining component is orthogonal to $\mathbf{u}$:

$$(\mathbf{v}-\mathbf{p})\cdot\mathbf{u} = \mathbf{v}\cdot\mathbf{u} - \frac{\mathbf{v}\cdot\mathbf{u}}{|\mathbf{u}|^2}(\mathbf{u}\cdot\mathbf{u}) = 0.$$



Example 1.4 (Lecture Example 4: angle and projection)**.** Let $\mathbf{a} = \langle 1, 2, 1\rangle$ and $\mathbf{b} = \langle -2, 1, 2\rangle$. Then

$$\mathbf{a}\cdot\mathbf{b} = -2 + 2 + 2 = 2, \quad |\mathbf{a}| = \sqrt{6}, \quad |\mathbf{b}| = 3.$$

Thus

$$\cos\theta = \frac{2}{3\sqrt{6}} = \frac{\sqrt{6}}{9}, \qquad \theta = \arccos\left(\frac{\sqrt{6}}{9}\right).$$

The projection of $\mathbf{a}$ onto $\mathbf{b}$ is

$$\operatorname{proj}_{\mathbf{b}} \mathbf{a} = \frac{2}{9}\langle -2, 1, 2\rangle = \left\langle -\frac{4}{9}, \frac{2}{9}, \frac{4}{9}\right\rangle.$$

1.5 The vector or cross product

The motivation is torque. A force $\mathbf{F}$ applied at position $\mathbf{r}$ relative to a fixed point O produces torque of magnitude $|\mathbf{r}||\mathbf{F}|\sin\theta$. Its axis is perpendicular to both $\mathbf{r}$ and $\mathbf{F}$, with the sense determined by a right-hand rule. This motivates the torque vector $\boldsymbol{\tau} = \mathbf{r}\times\mathbf{F}$. Viewed from the tip of $\boldsymbol{\tau}$ towards the origin, the induced rotation is anticlockwise.

Definition 1.1 (Cross product)**.** For nonzero, noncollinear vectors $\mathbf{a}, \mathbf{b}$, their cross product $\mathbf{c} = \mathbf{a}\times\mathbf{b}$ is characterised by

1. $|\mathbf{c}| = |\mathbf{a}||\mathbf{b}|\sin\theta$;
2. $\mathbf{c}$ is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$;
3. $\mathbf{a}, \mathbf{b}, \mathbf{c}$ form a right-handed triple.

If either vector is zero or the vectors are collinear, set their cross product to zero. Nonzero collinear vectors are parallel or antiparallel.

1.5.1 Deriving the Cartesian formula

Write $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$ and $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$. A perpendicular vector $\mathbf{c} = \langle x, y, z \rangle$ satisfies

$$a_1 x + a_2 y + a_3 z = 0, \qquad b_1 x + b_2 y + b_3 z = 0.$$

If $D = a_1 b_2 - a_2 b_1 \neq 0$, solving these equations gives

$$x = \frac{a_2 b_3 - a_3 b_2}{D} z, \qquad y = \frac{a_3 b_1 - a_1 b_3}{D} z.$$

Thus the perpendicular direction is

$$\mathbf{q} = \langle a_2 b_3 - a_3 b_2,\ a_3 b_1 - a_1 b_3,\ a_1 b_2 - a_2 b_1 \rangle, \qquad \mathbf{c} = t\mathbf{q}.$$

For noncollinear vectors at least one such minor is nonzero; if $D = 0$, choose another pair of coordinates. The component expression for $\mathbf{q}$ remains the same. Expanding and collecting terms gives

$$\begin{aligned} |\mathbf{q}|^2 &= (a_2 b_3 - a_3 b_2)^2 + (a_3 b_1 - a_1 b_3)^2 + (a_1 b_2 - a_2 b_1)^2 \\ &= (a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2) - (a_1 b_1 + a_2 b_2 + a_3 b_3)^2 \\ &= |\mathbf{a}|^2 |\mathbf{b}|^2 - (\mathbf{a} \cdot \mathbf{b})^2 \\ &= |\mathbf{a}|^2 |\mathbf{b}|^2 \sin^2 \theta. \end{aligned}$$

The magnitude condition on $\mathbf{c} = t\mathbf{q}$ gives $t = \pm 1$. The right-handed convention chooses $t = 1$, consistently with $\mathbf{i} \times \mathbf{j} = \mathbf{k}$. Therefore

$$\mathbf{a} \times \mathbf{b} = (a_2 b_3 - a_3 b_2)\mathbf{i} + (a_3 b_1 - a_1 b_3)\mathbf{j} + (a_1 b_2 - a_2 b_1)\mathbf{k}.$$

1.5.2 Properties and areas

The definition and component formula imply

$$\begin{aligned} \mathbf{a} \cdot (\mathbf{a} \times \mathbf{b}) &= 0, \\ \mathbf{a} \times \mathbf{b} &= -\mathbf{b} \times \mathbf{a}, \\ (\mathbf{a} + \mathbf{b}) \times \mathbf{c} &= \mathbf{a} \times \mathbf{c} + \mathbf{b} \times \mathbf{c}, \\ |\mathbf{a} \times \mathbf{b}|^2 + (\mathbf{a} \cdot \mathbf{b})^2 &= |\mathbf{a}|^2 |\mathbf{b}|^2. \end{aligned}$$

In particular,

$$\mathbf{i} \times \mathbf{j} = \mathbf{k}, \quad \mathbf{j} \times \mathbf{k} = \mathbf{i}, \quad \mathbf{k} \times \mathbf{i} = \mathbf{j}, \qquad \mathbf{i} \times \mathbf{i} = \mathbf{j} \times \mathbf{j} = \mathbf{k} \times \mathbf{k} = \mathbf{0}.$$

A parallelogram with adjacent edges $\mathbf{a}, \mathbf{b}$ has height $|\mathbf{b}| \sin \theta$ above a base of length $|\mathbf{a}|$, so

$$\text{Area(parallelogram)} = |\mathbf{a} \times \mathbf{b}|, \qquad \text{Area(triangle)} = \frac{1}{2} |\mathbf{a} \times \mathbf{b}|.$$

1.6 Triple products

The *scalar triple product* is

$$[\mathbf{a}, \mathbf{b}, \mathbf{c}] = \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}).$$

For a parallelepiped with edges $\mathbf{a}, \mathbf{b}, \mathbf{c}$, take the base to be the parallelogram spanned by $\mathbf{b}, \mathbf{c}$. For a nondegenerate base its area is $|\mathbf{b} \times \mathbf{c}|$. The height is the magnitude of the projection of $\mathbf{a}$ onto the normal:

$$\begin{aligned} h &= \left| \frac{\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})}{|\mathbf{b} \times \mathbf{c}|^2} (\mathbf{b} \times \mathbf{c}) \right| \\ &= \frac{|\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|}{|\mathbf{b} \times \mathbf{c}|}. \end{aligned}$$

Hence the volume is

$$V = |\mathbf{b} \times \mathbf{c}| h = \boxed{|[\mathbf{a}, \mathbf{b}, \mathbf{c}]|}.$$

This volume formula also gives zero for a degenerate parallelepiped. The scalar triple product is signed volume: it is positive for a right-handed triple and negative for a left-handed triple. It is unchanged by cyclic permutations and changes sign when any pair is swapped:

$$\begin{aligned} &[\mathbf{a}, \mathbf{b}, \mathbf{c}] = [\mathbf{b}, \mathbf{c}, \mathbf{a}] = [\mathbf{c}, \mathbf{a}, \mathbf{b}], \\ &[\mathbf{a}, \mathbf{b}, \mathbf{c}] = -[\mathbf{b}, \mathbf{a}, \mathbf{c}]. \end{aligned}$$

The *vector triple product* $\mathbf{a} \times (\mathbf{b} \times \mathbf{c})$ will be evaluated using index notation below.

1.7 Index notation

Relabel the standard basis as $\mathbf{e}_1 = \mathbf{i}$, $\mathbf{e}_2 = \mathbf{j}$, $\mathbf{e}_3 = \mathbf{k}$. **Einstein notation** convention suppresses the summation sign for a repeated index: here every summed index runs from 1 to 3. For example,

$$\mathbf{a} = a_i \mathbf{e}_i = \sum_{i=1}^{3} a_i \mathbf{e}_i.$$

A free index is not summed; a repeated dummy index can be renamed consistently.

1.7.1 The Kronecker delta and the dot product

Define

$$\delta_{ij} = \mathbf{e}_i \cdot \mathbf{e}_j = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

Then

$$\begin{aligned} \mathbf{a} \cdot \mathbf{b} &= (a_i \mathbf{e}_i) \cdot (b_j \mathbf{e}_j) \\ &= a_i b_j (\mathbf{e}_i \cdot \mathbf{e}_j) = \delta_{ij} a_i b_j = a_i b_i. \end{aligned}$$

The useful contraction rules are $\delta_{ij} A_j = A_i$ and $\delta_{ij} A_i = A_j$.

1.7.2 The Levi–Civita symbol and the cross product

Expand a cross product of basis vectors in the same basis:

$$\mathbf{e}_i \times \mathbf{e}_j = \varepsilon_{ijk} \mathbf{e}_k, \qquad \varepsilon_{ijk} = (\mathbf{e}_i \times \mathbf{e}_j) \cdot \mathbf{e}_k = [\mathbf{e}_i, \mathbf{e}_j, \mathbf{e}_k].$$

The last equality follows from the cyclic property of the scalar triple product. Since the basis is right-handed and orthonormal, $\varepsilon_{123} = 1$ and

$$\varepsilon_{ijk} = \begin{cases} +1, & (i,j,k) = (1,2,3), (2,3,1), (3,1,2), \\ -1, & (i,j,k) = (1,3,2), (3,2,1), (2,1,3), \\ 0, & \text{any two indices coincide.} \end{cases}$$

Thus $\varepsilon_{ijk} = \varepsilon_{jki} = \varepsilon_{kij}$ and swapping any two indices changes its sign. The cross product becomes

$$\boxed{\mathbf{a} \times \mathbf{b} = a_i b_j (\mathbf{e}_i \times \mathbf{e}_j) = \varepsilon_{ijk} a_i b_j \mathbf{e}_k}.$$

1.7.3 Contracting two Levi–Civita symbols

The identity needed for a vector triple product is

$$\boxed{\varepsilon_{kij}\varepsilon_{kn\ell} = \delta_{in}\delta_{j\ell} - \delta_{i\ell}\delta_{jn}}.$$

There is a sum over k. A nonzero term requires $i \neq j$, $n \neq \ell$ and a k different from all four indices. Consequently there are two nonzero cases: $i = n \neq j = \ell$ gives $+1$, while $i = \ell \neq j = n$ gives -1. In every other case the sum is zero, which is exactly the right-hand side.

1.7.4 The vector triple product

$$\begin{aligned} \mathbf{a} \times (\mathbf{b} \times \mathbf{c}) &= \mathbf{a} \times (\varepsilon_{ijk} b_i c_j \mathbf{e}_k) \\ &= \varepsilon_{ijk} b_i c_j a_\ell (\mathbf{e}_\ell \times \mathbf{e}_k) \\ &= \varepsilon_{ijk}\varepsilon_{\ell kn} a_\ell b_i c_j \mathbf{e}_n \\ &= \varepsilon_{kij}\varepsilon_{kn\ell} a_\ell b_i c_j \mathbf{e}_n \\ &= (\delta_{in}\delta_{j\ell} - \delta_{i\ell}\delta_{jn}) a_\ell b_i c_j \mathbf{e}_n \\ &= (a_j c_j) b_n \mathbf{e}_n - (a_i b_i) c_n \mathbf{e}_n \\ &= (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}. \end{aligned}$$

Therefore

$$\boxed{\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}}.$$

Week 2 ends with this calculation unfinished; Week 3 supplies the contraction identity and completes it. The recap is combined here rather than duplicated.

1.8 A determinant mnemonic and a plane example

The component formula can be remembered by expanding formally along the first row:

$$\begin{aligned} \mathbf{a} \times \mathbf{b} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} \\ &= \mathbf{i} \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} - \mathbf{j} \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} + \mathbf{k} \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix}, \end{aligned}$$

where $\left|\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right| = ad - bc$. The vector-valued first row is a mnemonic for cofactor expansion.

Example 1.5 (Lecture Example 5: plane through three points and distance)**.** Let

$$A=(-1,0,1),\qquad B=(2,-2,1),\qquad C=(3,-3,0),\qquad D=(-3,-1,2).$$

Find a Cartesian equation of the plane through A,B,C and its distance from D. A normal vector is

$$\begin{aligned}\mathbf{n}&=\overrightarrow{AB}\times\overrightarrow{AC}=\langle 3,-2,0\rangle\times\langle 4,-3,-1\rangle\\&=\mathbf{i}\big((-2)(-1)-0(-3)\big)-\mathbf{j}\big(3(-1)-0(4)\big)+\mathbf{k}\big(3(-3)-(-2)(4)\big)\\&=2\mathbf{i}+3\mathbf{j}-\mathbf{k}=\langle 2,3,-1\rangle.\end{aligned}$$

With $\mathbf{r}_A=\langle -1,0,1\rangle$, $\mathbf{r}_A\cdot\mathbf{n}=-3$. Thus the plane is

$$\boxed{2x+3y-z=-3}.$$

The shortest distance is the length of the projection of $\overrightarrow{AD}$ onto the normal. Since $\overrightarrow{AD}=\langle -2,-1,1\rangle$,

$$\begin{aligned}d&=\left|\operatorname{proj}_{\mathbf{n}}\overrightarrow{AD}\right|=\frac{|\overrightarrow{AD}\cdot\mathbf{n}|}{|\mathbf{n}|}\\&=\frac{|-4-3-1|}{\sqrt{4+9+1}}=\frac{8}{\sqrt{14}}=\boxed{\frac{4\sqrt{14}}{7}}.\end{aligned}$$

Chapter 2

Complex Numbers

This chapter follows the lecturer's algebra and geometry of complex numbers, which provide a foundation for later complex analysis. The examples and their calculations are retained in lecture order. Conditions on division and arguments are made explicit where needed.

2.1 Introduction and arithmetic

One motivation comes from physics and engineering. An alternating current

$$I(t) = A\cos(\omega t + \phi) = \operatorname{Re}(Ae^{i\omega t + i\phi})$$

can be handled conveniently using complex exponentials. Another motivation is polynomial equations: $x^2 + 1 = 0$ has no real solution, but it has the complex solutions $x = \pm i$.

Adjoin an imaginary unit i with $i^2 = -1$. It follows that

$$i^3 = -i, \qquad i^4 = 1,$$

and these powers repeat cyclically. A complex number has the form

$$z = x + iy, \qquad x, y \in \mathbb{R}.$$

Its real part is $\operatorname{Re} z = x$ and its imaginary part is $\operatorname{Im} z = y$; the imaginary part is a real number, not iy. The set of all complex numbers is

$$\mathbb{C} = \{x + iy : x, y \in \mathbb{R}\}.$$

Addition and multiplication use the usual algebraic rules with $i^2 = -1$:

$$\begin{aligned}(x_1 + iy_1) + (x_2 + iy_2) &= (x_1 + x_2) + i(y_1 + y_2),\\ (x_1 + iy_1)(x_2 + iy_2) &= x_1x_2 + ix_1y_2 + ix_2y_1 + i^2y_1y_2\\ &= (x_1x_2 - y_1y_2) + i(x_1y_2 + x_2y_1).\end{aligned}$$

For $z = x + iy \neq 0$, rationalise the denominator to find its inverse:

$$\begin{aligned}z^{-1} = \frac{1}{x+iy} &= \frac{x - iy}{(x+iy)(x-iy)}\\ &= \frac{x-iy}{x^2+y^2} = \frac{x}{x^2+y^2} - i\frac{y}{x^2+y^2}.\end{aligned}$$

The denominator is nonzero because x and y are not both zero. These operations make $\mathbb{C}$ an algebraic field; division is defined for every nonzero denominator.

Example 2.1 (Lecture Example 1: a reciprocal)**.** For $z = 1 + i$,

$$\frac{1}{z} = \frac{1}{1+i}\frac{1-i}{1-i} = \frac{1-i}{(1+i)(1-i)}$$
$$= \frac{1-i}{1-i^2} = \frac{1-i}{2} = \frac{1}{2} - \frac{1}{2}i.$$

2.1.1 Conjugate and modulus

For $z = x + iy$, define its conjugate and modulus by

$$\overline{z} = x - iy, \qquad |z| = \sqrt{x^2 + y^2} \in [0, \infty).$$

Then

$$z\overline{z} = (x + iy)(x - iy) = x^2 + y^2 = |z|^2.$$

Thus $|z| = \sqrt{z\overline{z}}$ and, for $z \neq 0$,

$$\boxed{\frac{1}{z} = \frac{\overline{z}}{z\overline{z}} = \frac{\overline{z}}{|z|^2}}.$$

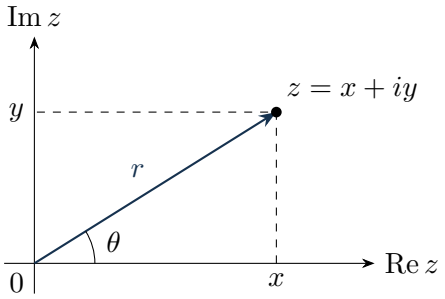
2.2 Polar coordinates and powers

A complex number $z = x + iy$ corresponds to the point (x, y) in the complex (or Argand) plane. For $z \neq 0$, write

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad r = \sqrt{x^2 + y^2} = |z| > 0.$$

The angle θ is an argument of z; adding an integer multiple of 2π gives another argument. One may choose $0 \leq \theta < 2\pi$, or use another interval of length 2π consistently. The argument of 0 is undefined. The polar form is

$$z = r(\cos\theta + i\sin\theta).$$



2.2.1 Multiplication in polar form

Let $z_j = r_j(\cos\theta_j + i\sin\theta_j)$ for $j = 1, 2$. Using the trigonometric addition formulas,

$$z_1 z_2 = r_1 r_2\big[\cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2$$
$$+ i(\sin\theta_1\cos\theta_2 + \sin\theta_2\cos\theta_1)\big]$$
$$= r_1 r_2\big[\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2)\big].$$

The moduli multiply and the arguments add, with arguments understood modulo 2π. In particular,

$$(\cos\theta + i\sin\theta)^2 = \cos 2\theta + i\sin 2\theta,$$
$$(\cos\theta + i\sin\theta)^3 = \cos 3\theta + i\sin 3\theta.$$

Induction gives the same rule for every positive integer power. It also holds for power zero. For $m > 0$, taking a reciprocal gives

$$\begin{aligned}(\cos\theta + i\sin\theta)^{-m} &= \frac{1}{\cos(m\theta) + i\sin(m\theta)} \\ &= \frac{\cos(m\theta) - i\sin(m\theta)}{\cos^2(m\theta) + \sin^2(m\theta)} \\ &= \cos(-m\theta) + i\sin(-m\theta),\end{aligned}$$

since cosine is even and sine is odd. Therefore

Theorem 2.1 (de Moivre's theorem)**.** *For every integer n and real θ,*

$$\boxed{(\cos\theta + i\sin\theta)^n = \cos(n\theta) + i\sin(n\theta)}.$$

Example 2.2 (Lecture Example 2: a multiple-angle polynomial)**.** Express $\sin 5\theta / \sin\theta$ as a polynomial in $\cos\theta$ where $\sin\theta \neq 0$. Put $c = \cos\theta$ and $s = \sin\theta$. By de Moivre and the binomial formula,

$$\begin{aligned}\cos 5\theta + i\sin 5\theta &= (c + is)^5 \\ &= c^5 + 5c^4 is + 10c^3 i^2 s^2 + 10c^2 i^3 s^3 + 5ci^4 s^4 + i^5 s^5 \\ &= (c^5 - 10c^3 s^2 + 5cs^4) + i(5c^4 s - 10c^2 s^3 + s^5).\end{aligned}$$

Equating imaginary parts and dividing by $s \neq 0$,

$$\begin{aligned}\frac{\sin 5\theta}{\sin\theta} &= 5c^4 - 10c^2 s^2 + s^4 \\ &= 5c^4 - 10c^2(1 - c^2) + (1 - c^2)^2 \\ &= \boxed{16\cos^4\theta - 12\cos^2\theta + 1}.\end{aligned}$$

2.3 The complex exponential and Euler's formula

Euler's formula is

$$\boxed{e^{i\theta} = \cos\theta + i\sin\theta}.$$

The lecture notes its connection with the power-series representations of e^x, sine and cosine; these series are introduced in Chapter 3 and studied further in analysis. Replacing θ by $-\theta$ gives

$$e^{-i\theta} = \cos\theta - i\sin\theta.$$

Adding and subtracting these identities gives

$$\boxed{\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}}, \qquad \boxed{\sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}}.$$

Polar form can now be written $z = re^{i\theta}$. In particular, $e^{i\pi} = -1$ and $e^{2\pi i} = 1$.

Example 2.3 (Lecture Example 3: expanding a power of cosine)**.** Show that

$$\cos^6\theta = \frac{1}{32}\cos 6\theta + \frac{3}{16}\cos 4\theta + \frac{15}{32}\cos 2\theta + \frac{5}{16}.$$

Let $z = e^{i\theta}$. Then $e^{-i\theta} = z^{-1} = \overline{z}$ and $\cos\theta = (z + z^{-1})/2$. Consequently,

$$\begin{aligned}\cos^6\theta &= \frac{1}{64}(z + z^{-1})^6 \\ &= \frac{1}{64}(z^6 + 6z^4 + 15z^2 + 20 + 15z^{-2} + 6z^{-4} + z^{-6}) \\ &= \frac{1}{32}\frac{z^6 + z^{-6}}{2} + \frac{3}{16}\frac{z^4 + z^{-4}}{2} + \frac{15}{32}\frac{z^2 + z^{-2}}{2} + \frac{5}{16} \\ &= \frac{1}{32}\cos 6\theta + \frac{3}{16}\cos 4\theta + \frac{15}{32}\cos 2\theta + \frac{5}{16}.\end{aligned}$$

2.4 Roots of complex numbers

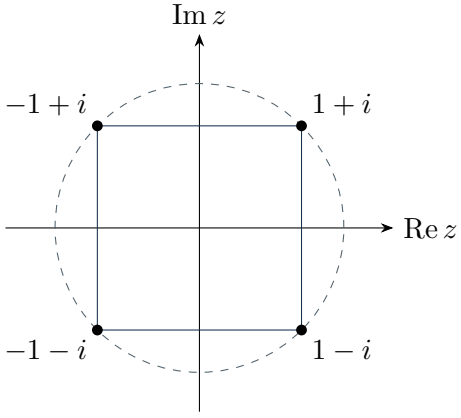
Example 2.4 (Lecture Example 4: the fourth roots of -4)**.** Find all $z \in \mathbb{C}$ such that $z^4 = -4$. Since $-4 = 4e^{i\pi}$, write $z = re^{i\theta}$ with $r > 0$. Then

$$r^4 e^{4i\theta} = 4e^{i\pi} \quad \Longleftrightarrow \quad r^4 = 4, \qquad 4\theta = \pi + 2\pi k \quad (k \in \mathbb{Z}).$$

Thus $r = \sqrt{2}$ and $\theta = \pi/4 + k\pi/2$. Four consecutive values of k give all distinct roots; further values repeat them. Taking $k = 0, 1, 2, 3$,

$$\begin{aligned}z_0 &= \sqrt{2}e^{i\pi/4} = 1 + i, \\ z_1 &= \sqrt{2}e^{3i\pi/4} = -1 + i, \\ z_2 &= \sqrt{2}e^{5i\pi/4} = -1 - i, \\ z_3 &= \sqrt{2}e^{7i\pi/4} = 1 - i.\end{aligned}$$

Negative arguments, as used for two of the roots in the manuscript, give the same numbers: $e^{5i\pi/4} = e^{-3i\pi/4}$ and $e^{7i\pi/4} = e^{-i\pi/4}$.



2.5 The geometry of complex numbers

Example 2.5 (Lecture Example 5: a modulus locus)**.** Given a, find all $z \in \mathbb{C}$ such that

$$\left|\frac{z-1}{z+3}\right| = a, \qquad z \neq -3.$$

The left-hand side is real and nonnegative. Hence there are no solutions if $a \notin \mathbb{R}$ or $a < 0$. If $a = 0$, the only solution is $z = 1$. For $a > 0$, put $z = x + iy$. Squaring gives

$$\begin{aligned} \frac{(x-1)^2 + y^2}{(x+3)^2 + y^2} &= a^2, \\ (x-1)^2 + y^2 &= a^2\big((x+3)^2 + y^2\big), \\ (1-a^2)(x^2+y^2) - 2(1+3a^2)x &= 9a^2 - 1. \end{aligned}$$

If $a = 1$, the last equation is $-8x = 8$, so the locus is the line

$$\boxed{x = -1}.$$

If $a > 0$ and $a \neq 1$, divide by $1 - a^2$ and complete the square:

$$\begin{aligned} x^2 + y^2 - \frac{2(1+3a^2)}{1-a^2}x &= \frac{9a^2-1}{1-a^2}, \\ \left(x - \frac{1+3a^2}{1-a^2}\right)^2 + y^2 &= \frac{9a^2-1}{1-a^2} + \frac{(1+3a^2)^2}{(1-a^2)^2} \\ &= \frac{16a^2}{(1-a^2)^2}. \end{aligned}$$

Thus the locus is a circle with centre and radius

$$\boxed{\left(\frac{1+3a^2}{1-a^2}, 0\right)}, \qquad \boxed{R = \frac{4a}{|1-a^2|}}.$$

The excluded point $z = -3$ is not on any of these solution loci. Geometrically, the original condition prescribes the ratio of the distances from z to 1 and to -3.

My questions and revision notes

Chapter 3

Differential Calculus

This chapter concerns functions of one real variable. It follows the lecturer's six topics, including the introduction to integration and series approximations. The examples retain their lecture order; selected lecture diagrams are redrawn schematically. Conditions that were implicit in the manuscript are stated explicitly. Throughout this chapter, log means the natural logarithm and angles are in radians.

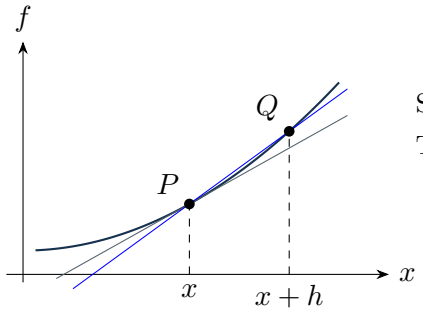
3.1 The derivative

Definition 3.1 (Differentiability)**.** Let $(a, b) \subseteq \mathbb{R}$ and $f : (a, b) \to \mathbb{R}$. The function f is differentiable at $x \in (a, b)$ if the finite limit

$$f'(x) = \lim_{h \to 0} \frac{f(x + h) - f(x)}{h} \tag{1}$$

exists. Other notation is $\mathrm{d}f/\mathrm{d}x$. The function is differentiable on (a, b) if this derivative exists at every point of the interval.

Geometrically, $f'(x)$ is the slope of the tangent to the graph at x. The difference quotient is the slope of the secant through the points with horizontal coordinates x and $x+h$; the tangent slope is its limit. In applications a derivative measures an instantaneous rate of change. For example, if $x(t)$ is position at time t, then $x'(t)$ is velocity. A rigorous treatment of differentiation is developed in Analysis 1.



Example 3.1 (Lecture Example 1: use the definition)**.** Find the derivative of $f(x) = x^2$ directly

from (1).

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{(x+h)^2 - x^2}{h} \\ &= \lim_{h\to 0} \frac{x^2 + 2xh + h^2 - x^2}{h} \\ &= \lim_{h\to 0} (2x + h) = 2x. \end{aligned}$$

3.1.1 Basic derivative formulas

$f(x)$	$f'(x)$	Domain used here
x^α	$\alpha x^{\alpha-1}$	$x > 0$, $\alpha \in \mathbb{R}$
e^x	e^x	$x \in \mathbb{R}$
$\log x$	$1/x$	$x > 0$
$\sin x$	$\cos x$	$x \in \mathbb{R}$
$\cos x$	$-\sin x$	$x \in \mathbb{R}$

For integer powers the power rule also applies at other points where the function and the displayed derivative are defined.

3.1.2 Rules of differentiation

If f and g are differentiable and λ is constant, then

$$(f+g)' = f' + g', \qquad (\lambda f)' = \lambda f',$$
$$(fg)' = f'g + fg'.$$

The first two identities express linearity of differentiation. For a composition, the chain rule is

$$\frac{\mathrm{d}}{\mathrm{d}x} g(f(x)) = g'(f(x))f'(x), \qquad (g \circ f)' = (g' \circ f)f'.$$

These rules follow from the derivative definition using the limit tools of analysis.

3.2 Inverse functions and their derivatives

If f is strictly monotone on an interval, it is injective there. In the increasing case, $x_1 > x_2$ implies $f(x_1) > f(x_2)$; in the decreasing case, $x_1 > x_2$ implies $f(x_1) < f(x_2)$. Taking the codomain to be its image, f therefore has an inverse:

$$y = f(x) \quad \Longleftrightarrow \quad x = f^{-1}(y).$$

Write $g = f^{-1}$. Then $g(f(x)) = x$. For a differentiable, strictly monotone f, at points where $f'(x) \neq 0$, the inverse is differentiable and the chain rule gives

$$g'(f(x))f'(x) = 1.$$

Substituting $x = f^{-1}(y)$ yields

$$g'(y) = \frac{1}{f'(f^{-1}(y))}. \tag{2}$$

In function notation,

$$(f^{-1})' = \frac{1}{f' \circ f^{-1}}. \tag{2'}$$

Remark. The nonzero-derivative condition is essential for this finite derivative formula. The notation f^{-1} here means an inverse function, not $1/f$.

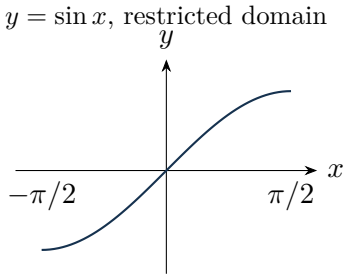
Example 3.2 (Lecture Example 2: the inverse sine)**.** Find the derivative of $\arcsin x = \sin^{-1} x$ for $-1 < x < 1$. Restrict sine to $(-\pi/2, \pi/2)$, on which it is strictly increasing and has image $(-1, 1)$. Formula (2) gives

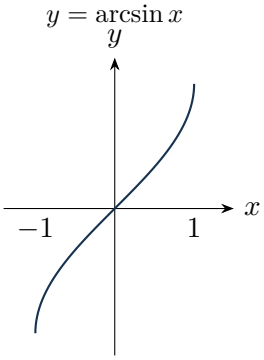
$$(\arcsin x)' = \frac{1}{\cos(\arcsin x)}.$$

Put $\theta = \arcsin x$. Then $\sin\theta = x$ and $-\pi/2 < \theta < \pi/2$, so $\cos\theta > 0$. Consequently,

$$\cos(\arcsin x) = \sqrt{1 - \sin^2\theta} = \sqrt{1 - x^2}, \qquad \boxed{(\arcsin x)' = \frac{1}{\sqrt{1 - x^2}}}.$$

The positive square root follows from the chosen range of arcsin.





3.3 Implicit differentiation

The inverse-function discussion concerns cases where one variable can be expressed in terms of the other. More generally, a relation between x and y may define a curve without a convenient explicit formula $y = f(x)$. To find its slope locally, regard y as a differentiable function $y(x)$ and differentiate the relation with respect to x. Apply the chain rule whenever differentiating an expression involving $y(x)$.

Example 3.3 (Lecture Example 3: slope of an implicit curve)**.** Find the slope at $(x, y) = (2, 1)$ of the curve

$$y^7 + x^4y^3 - 10x + 3 = 0.$$

First, $1 + 16 - 20 + 3 = 0$, so the point lies on the curve. Differentiating with respect to x gives

$$\begin{aligned} 0 &= \frac{\mathrm{d}}{\mathrm{d}x}\left(y(x)^7 + x^4y(x)^3 - 10x + 3\right) \\ &= 7y^6y' + 4x^3y^3 + 3x^4y^2y' - 10. \end{aligned}$$

The first term uses the chain rule. Differentiating $x^4y(x)^3$ uses both the product rule and the chain rule. Collect the terms involving y':

$$\begin{aligned} y'(7y^6 + 3x^4y^2) &= 10 - 4x^3y^3, \\ y' &= \frac{10 - 4x^3y^3}{7y^6 + 3x^4y^2}. \end{aligned}$$

At $(2, 1)$ the denominator is $55 \neq 0$, and

$$\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{(2,1)} = \frac{10 - 4 \cdot 2^3 \cdot 1^3}{7 \cdot 1^6 + 3 \cdot 2^4 \cdot 1^2} = -\frac{22}{55} = -\frac{2}{5}.$$

Example 3.4 (Lecture Example 4: a self-exponential)**.** For $x > 0$, find the derivative of $f(x) = x^x$. Let $y = x^x$. Taking natural logarithms gives $\log y = x \log x$. Differentiating this identity,

$$\frac{y'}{y} = \log x + x\frac{1}{x} = \log x + 1,$$
$$y' = y(\log x + 1).$$

Therefore

$$\boxed{f'(x) = x^x(\log x + 1)}.$$

3.4 Applications of derivatives

3.4.1 Optimisation: maxima and minima

Theorem 3.2 (Necessary condition for an interior extremum)**.** *Let f be differentiable on (a, b) and let $c \in (a, b)$. If f has a local maximum or minimum at c, then $f'(c) = 0$.*

Proof sketch from the lecture. Suppose f has a local maximum at c. For sufficiently small h, $f(c + h) - f(c) \leq 0$. Hence

$$\frac{f(c+h) - f(c)}{h} \leq 0 \quad (h > 0), \qquad \frac{f(c+h) - f(c)}{h} \geq 0 \quad (h < 0).$$

Differentiability means that the two one-sided limits exist and agree. The first inequality implies $f'(c) \leq 0$ and the second implies $f'(c) \geq 0$. Thus $f'(c) = 0$. The minimum case is analogous. □

To find an absolute maximum or minimum of a differentiable function on a closed interval, find interior stationary points and compare the values there with the endpoint values. For continuous functions that fail to be differentiable at some interior points, those points must also be checked. A stationary point alone does not guarantee an extremum.

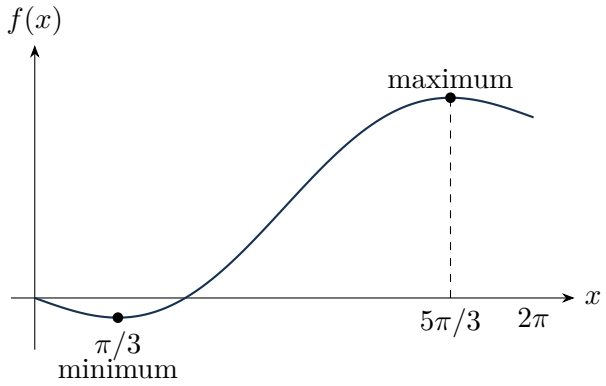
Example 3.5 (Lecture Example 5: include the endpoints)**.** Find the maximum and minimum of $f(x) = x - 2\sin x$ on $[0, 2\pi]$.

$$f'(x) = 1 - 2\cos x,$$
$$f'(x) = 0 \quad \Longleftrightarrow \quad \cos x = \frac{1}{2}.$$

The interior candidates are $x = \pi/3$ and $x = 5\pi/3$. Their values and the endpoint values are

$$f(\pi/3) = \frac{\pi}{3} - \sqrt{3} \approx -0.685,$$
$$f(5\pi/3) = \frac{5\pi}{3} + \sqrt{3} \approx 6.968,$$
$$f(0) = 0, \qquad f(2\pi) = 2\pi \approx 6.283.$$

Thus the absolute minimum is $\pi/3 - \sqrt{3}$ at $x = \pi/3$, and the absolute maximum is $5\pi/3 + \sqrt{3}$ at $x = 5\pi/3$.



3.4.2 First-order approximation

The tangent line to f at $x = c$ has equation

$$f_{\mathrm{t}}(x) - f(c) = f'(c)(x - c).$$

For x close to c, this gives the first-order approximation

$$\boxed{f(x) \approx f(c) + f'(c)(x - c)}.$$

Example 3.6 (Lecture Example 6: estimate $\sqrt{17}$)**.** Take $f(x) = \sqrt{x}$ and $c = 16$. Then

$$f(16) = 4, \qquad f'(x) = \frac{1}{2\sqrt{x}}, \qquad f'(16) = \frac{1}{8}.$$

Consequently,

$$\sqrt{17} = f(17) \approx 4 + \frac{1}{8}(17 - 16) = 4.125.$$

A calculator gives $\sqrt{17} \approx 4.123106$, so the absolute error is about 0.0019, or approximately 0.002 as noted in the lecture.

3.5 Integration

Geometrically, a definite integral gives signed area under a graph. In applications it represents accumulation of a changing quantity: integrating velocity gives displacement; integrating speed gives distance travelled. The lecture introduces integration through areas of piecewise constant approximations; a rigorous treatment belongs to Analysis 2.

3.5.1 Partitions and sums of rectangle areas

Let $f : [a, b] \to \mathbb{R}$ be continuous and take a partition

$$a = x_0 < x_1 < \cdots < x_{n-1} < x_n = b.$$

A right-endpoint rectangle on $[x_{i-1}, x_i]$ has signed area $f(x_i)(x_i - x_{i-1})$. Adding the areas gives a Riemann sum. For equal-width intervals, put

$$\Delta x = \frac{b - a}{n}, \qquad x_i = a + i\Delta x.$$

The definite integral is then

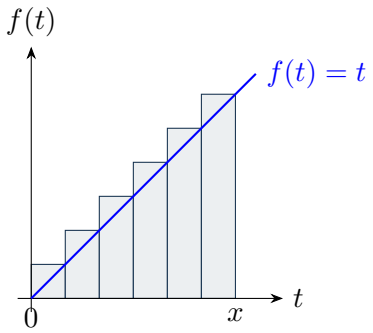
$$\boxed{\int_a^b f(x)\,\mathrm{d}x = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\Delta x}.$$

For arbitrary partitions, the limiting requirement is that the largest subinterval width tends to zero; merely increasing the number of intervals is not enough. The continuity assumption ensures the sums converge to the same value for these choices.

Example 3.7 (Lecture Example 7: integrate directly from the definition)**.** For $f(t) = t$ and $x > 0$, find $\int_0^x f(t)\,\mathrm{d}t$. Divide $[0, x]$ into n equal intervals with endpoints $t_i = ix/n$.

$$\begin{aligned}\int_0^x t\,\mathrm{d}t &= \lim_{n\to\infty} \sum_{i=1}^{n} \left(\frac{ix}{n}\right)\frac{x}{n} \\ &= \lim_{n\to\infty} \frac{x^2}{n^2} \sum_{i=1}^{n} i \\ &= \lim_{n\to\infty} \frac{x^2}{n^2}\frac{n(n+1)}{2} \\ &= x^2 \lim_{n\to\infty} \frac{n(n+1)}{2n^2} = \frac{x^2}{2}.\end{aligned}$$

The identity $\sum_{i=1}^{n} i = n(n+1)/2$ is used in the third line.



Right-endpoint rectangles

3.5.2 The fundamental theorem of calculus

In the preceding example,

$$F(x) = \int_0^x t\,\mathrm{d}t = \frac{x^2}{2} \quad \text{has} \quad F'(x) = x = f(x).$$

This relation holds more generally.

Theorem 3.3 (**Fundamental theorem of calculus**)**.** *Let f be continuous on $[a, b]$ and set*

$$F(x) = \int_a^x f(t)\,\mathrm{d}t.$$

Then $F'(x) = f(x)$ for $a < x < b$. Moreover, if G is any antiderivative of f, so $G' = f$, then

$$\boxed{\int_a^b f(t)\,\mathrm{d}t = G(b) - G(a)}.$$

This result is also called the Newton–Leibniz theorem.

Geometric proof idea. For small h, additivity of area gives

$$F(x+h) = F(x) + \int_x^{x+h} f(t)\,\mathrm{d}t.$$

Continuity makes the narrow strip approximately a rectangle of height $f(x)$ and signed width h, so

$$F(x+h) - F(x) \approx f(x)h, \qquad \frac{F(x+h)-F(x)}{h} \longrightarrow f(x).$$

This explains why differentiating accumulated area recovers the original function. The approximation here is a proof idea, rather than a replacement for the limit argument in Analysis 2.

3.5.3 Antiderivatives and indefinite integrals

An antiderivative (or primitive) of f is a function G satisfying $G' = f$. The notation

$$\int f(x)\,\mathrm{d}x = G(x) + C$$

represents the family of antiderivatives, which differ by an additive constant on an interval. For example,

$$\int x\,\mathrm{d}x = \frac{x^2}{2} + C.$$

A definite integral is a number; an indefinite integral describes functions. The fundamental theorem explains the sense in which integration reverses differentiation.

3.5.4 Properties and elementary formulas

For integrable f, g and a constant λ,

$$\begin{aligned}
\int_a^b (f+g)\,\mathrm{d}x &= \int_a^b f\,\mathrm{d}x + \int_a^b g\,\mathrm{d}x,\\
\int_a^b \lambda f\,\mathrm{d}x &= \lambda \int_a^b f\,\mathrm{d}x,\\
\int_a^b f\,\mathrm{d}x &= \int_a^c f\,\mathrm{d}x + \int_c^b f\,\mathrm{d}x \qquad (a < c < b).
\end{aligned}$$

The first two properties express linearity of integration.

$f(x)$	$\int f(x)\,\mathrm{d}x$	Condition
x^α	$\dfrac{x^{\alpha+1}}{\alpha+1} + C$	$x > 0$, $\alpha \neq -1$
e^x	$e^x + C$	$x \in \mathbb{R}$
$1/x$	$\log\lvert x\rvert + C$	an interval avoiding 0
$\sin x$	$-\cos x + C$	$x \in \mathbb{R}$
$\cos x$	$\sin x + C$	$x \in \mathbb{R}$

For a definite integral the same constant cancels: $(G(b)+C) - (G(a)+C) = G(b) - G(a)$.

3.6 Series approximation

3.6.1 From a tangent line to a quadratic

Recall the first-order approximation near $x = c$:

$$f(x) \approx f(c) + f'(c)(x - c).$$

To understand higher orders, first take the centre to be 0. The linear polynomial $P_1(x) = f(0) + f'(0)x$ matches both the value and the first derivative of f at 0. For a second-order approximation, try

$$P_2(x) = f(0) + f'(0)x + Ax^2.$$

Matching derivatives at 0 gives

$$\begin{aligned} P_2'(x) &= f'(0) + 2Ax, & P_2'(0) &= f'(0), \\ P_2''(x) &= 2A, & P_2''(0) &= 2A = f''(0). \end{aligned}$$

Thus $A = f''(0)/2$, and

$$f(x) \approx f(0) + f'(0)x + \frac{f''(0)}{2}x^2.$$

Exercise from handwritten p. 60. Find a second-order approximation to $\sqrt{17}$ and compare it with a calculator. The margin of the lecture records approximately 4.123046 for the quadratic approximation, compared with 4.123106 from a calculator.

3.6.2 Maclaurin and Taylor polynomials

Matching further derivatives extends the construction to degree N:

$$P_{N,0}(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots + \frac{f^{(N)}(0)}{N!}x^N.$$

The associated infinite series centred at 0 is the *Maclaurin series*:

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n. \tag{3}$$

For a general centre a, the degree-N Taylor polynomial is

$$P_{N,a}(x) = \sum_{n=0}^{N} \frac{f^{(n)}(a)}{n!}(x - a)^n,$$

and the associated *Taylor series* is

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x - a)^n. \tag{3'}$$

Here $f^{(0)} = f$ and $0! = 1$.

Remark (Approximation versus equality). The lecture introduces these expressions using an approximation sign. A finite Taylor polynomial approximates f near its centre under suitable smoothness assumptions. An infinite Taylor series equals $f(x)$ only where the remainder $f(x) - P_{N,a}(x)$ tends to zero as $N \to \infty$. The existence of derivatives of every order alone does not ensure this. This distinction will be treated more fully in analysis.

Example 3.8 (Lecture Example 8: the exponential function)**.** Let $f(x) = e^x$. Every derivative is e^x, so $f^{(n)}(0) = 1$ for all $n \geq 0$. Substituting in (3) gives

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots .$$

This series converges to e^x for every real x, so the approximation can be replaced by equality. The series can also be used as a definition of the exponential function.

3.6.3 Similar examples: sine and cosine

The corresponding expansions are

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots ,$$

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots .$$

Both equalities hold for all real x. In the lecture's expanded forms, $3! = 6$, $5! = 120$ and $4! = 24$.

Example 3.9.

$$\begin{aligned}
f(x) &= log(1+x), |x| < 1. \\
f(0) &= log1 = 0 \\
f'(x) &= \frac{1}{1+x} \to f'(0) = 1 \\
f''(x) &= -\frac{1}{(1+x)^2} = -(1+x)^{-2} \to f''(0) = -1 \\
f'''(x) &= 2(1+x)^{-3} \to f'''(0) = 2 \\
&.... \\
f^{(n)}(x) &= (-1)^{n-1}(n-1)!
\end{aligned}$$

Maclaurin's Series:

$$log(1+x) \approx \sum_{n=0}^{\infty} (-1)^{n-1} \frac{(n-1)!}{n!} x^n$$

$$\Rightarrow \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - ...$$

Example 3.10 (Binomial Expansion)**.**

$$f(x) = (1+x)^{\alpha}, \qquad \alpha \in \mathbb{R}.$$

For $|x| < 1$, its Maclaurin series is

$$\begin{aligned}
(1+x)^{\alpha} &= 1 + \sum_{n=1}^{\infty} \frac{\alpha(\alpha-1)\cdots(\alpha-n+1)}{n!} x^n \\
&= \sum_{n=0}^{\infty} \binom{\alpha}{n} x^n.
\end{aligned}$$

Here $\binom{\alpha}{0} = 1$.

NB: If α is a non-negative integer, the series terminates and the identity holds for all real x. Otherwise, convergence at $x = 1$ and $x = -1$ must be checked separately.

Example 3.11. Maclaurin Series o

Chapter 4

Partial Derivatives